PhysioFlow: an open-source interactive platform for crop ecophysiology and agronomic data analysis with statistical and experimental-design guard-rails

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Abstract

Field ecophysiology campaigns using infrared gas analyzers (IRGAs), chlorophyll meters, and ceptometers produce tabular datasets of gas-exchange, chlorophyll fluorescence, and leaf-area variables that must be cleaned, explored, modeled, and mapped before publication. Instrument vendors provide no free open-source software for these stages, forcing researchers to write ad hoc scripts. We present *PhysioFlow*, an open-source Python/Streamlit interactive platform that ingests CSV/TSV/XLSX exports, validates them against an explicit schema, and exposes a configurable replicate-handling pipeline, exploratory analysis with statistical guard-rails (categorical confounding via Cramér's V, VIF with derived-variable awareness, Kruskal–Wallis with minimum-N filtering, normality tests with companion Q–Q plots,

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multi-method outlier consensus), predictive modelling (regression and classification) with optional GroupKFold, geospatial analysis in UTM metres and STL time-series decomposition with sparsity guard-rails, in a single code-free trilingual (PT/EN/ES) interface. A dedicated module brings the classical agronomic designs into the same interface, from completely randomised and randomised complete block to factorial, Latin square, split-plot, strip-plot and nested layouts, together with ANCOVA and dose-response regression; each fit reports its assumption checks and a post-hoc mean comparison (Tukey, Scott–Knott, Duncan, LSD, Scheffé or Dunnett), and all of these were cross-validated against R. Because an automatic data-profile recognises whether the uploaded table is an ecophysiology dataset or an arbitrary one, the platform is no longer tied to ecophysiology and analyses any tabular agronomic or biological dataset. Released under GPL-2.0 with a mandatory citation clause, it is used in the Goiás Verde initiative.

Keywords: Crop ecophysiology, Photosynthesis, IRGA, Exploratory data analysis, Experimental design, Analysis of variance, Statistical guard-rails, Reproducible research

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Required Metadata

Current code version

Nr.	**Code metadata description**	**Value**
C1	Current code version	v1.0
C2	Permanent GitHub link to code/repository	`https://github.com/ML-Carbon-Project/physioFlow`
C3	Legal Code License	GPLv2 with mandatory citation clause
C4	Code versioning system used	git
C5	Software code languages, tools, services used	Python 3.10+; Streamlit; pandas, NumPy, SciPy, scikit-learn, statsmodels; geopandas, shapely, pyproj, libpysal, esda; pytest
C6	Compilation requirements, operating environments & dependencies	Linux, macOS, Windows; Python $\geq$ 3.10; `pip install -r requirements.txt`
C7	Link to developer documentation/manual	`https://github.com/ML-Carbon-Project/physioFlow/tree/main/docs`
C8	Support email for questions	`leandrorodrigues.s@gmail.com`

Table 1: Code metadata (mandatory).

1. Motivation and significance

Plant ecophysiology links the carbon, water and energy economy of crops to their environment, and its field measurement underpins decisions in agronomy, breeding and climate-change adaptation [1, 2]. Three classes of portable instruments dominate field campaigns. Infrared gas analyzers (IRGA) housed in leaf chambers quantify net photosynthesis (A), transpiration (E), stomatal conductance (g_s) and internal/ambient CO_2 (C_i, C_a), from which derived efficiencies such as water-use efficiency (WUE), carboxylation efficiency (A/C_i) and the C_i/C_a ratio are computed [3, 4, 5]. Chlorophyll fluorometers and chlorophyll-content meters report the effective quantum yield of photosystem II (Y_{II}), the electron transport rate (ETR) and chlorophyll a/b [6, 7]. Ceptometers estimate the leaf area index (LAI) [8]. Together they characterize the photosynthetic performance of a crop across genotypes, phenological stages, management systems and seasons. These measurements anchor agronomic studies of crop responses to irrigation and salinity, soil management and seasonal climate, in which leaf gas exchange and soil–plant carbon fluxes are read jointly to interpret productivity and stress [9, 10, 11, 12, 13]. In tropical agro-ecosystems such as the Brazilian Cerrado, where the present case study is set, characterizing these fluxes is central both to yield management and to ecosystem carbon-balance accounting [14].

Each instrument exports a wide CSV/XLSX table in which a row is a leaf or canopy measurement carrying gas-exchange variables, fluorescence variables, chlorophyll and LAI replicates, plus metadata (crop, farm, land use, season, sampling point, coordinates, date). Turning this raw output into a publishable analysis still demands several non-trivial steps: standardizing categorical fields; discarding rows without minimum agronomic information; consolidating field replicates of chlorophyll and LAI; running exploratory and inferential analyses; checking multicollinearity among the strongly derived physiological indices; inspecting the spatial structure of the campaign; decomposing temporal trends; and, optionally, fitting predictive models that relate measured traits to photosynthesis.

Despite the wide commercial availability of these instruments, manufacturers do not provide free, open-source software for the exploratory and modeling stages that transform raw exports into results. Vendor viewers focus on acquisition and quality control. Consequently, researchers must export the data and write custom scripts in R [15] or Python, which poses a considerable barrier for agronomists, plant scientists and graduate students without formal training in programming or statistics. In most laboratories, the intervening steps are still done by hand or are half-automated using spreadsheets and a handful of disconnected statistical packages. This is slow, easy to get wrong, and difficult for a colleague to reproduce; in a high-throughput campaign, it becomes the bottleneck that holds up the whole analysis. The structure of ecophysiology field data also invites silent statistical errors: physiological indices are mutually derived (e.g. $\mathrm{WUE} = A/E$), inflating multicollinearity by construction; sampling designs frequently alias farm, crop and land use, so an apparent "between-farm" difference may be an unidentifiable "between-crop" difference; replicated measurements at the same site induce pseudoreplication [16] that inflates model skill; and short campaigns offer too few dates for meaningful seasonal decomposition.

Inspired by recent open-source efforts that bridged data analysis and end users through a Streamlit interface [17], and by our sibling platform for chamber-based soil greenhouse-gas flux analysis [18], we developed *PhysioFlow* to consolidate the entire ecophysiology data-analysis workflow into a single configurable, interactive platform. The distinguishing design choice is a set of *statistical guard-rails* woven into the user interface: the application not only computes statistics but warns when their assumptions are violated. Recent software papers in neighbouring agronomic niches [19, 20] have shown how much is gained by releasing an instrument-bound, end-to-end analytical pipeline as citable software, and PhysioFlow does the same for crop ecophysiology.

2. Software description

2.1. Software architecture

PhysioFlow is implemented in Python 3.10 [17, 21, 22] and follows a *session-state-centric, page-renderer pattern* typical of Streamlit applications, in which each page is a self-contained module that reads from and writes to a shared `st.session_state` and consumes the cleaning, statistics and visualisation primitives in the `src/` package, so that costly I/O and cleaning operations happen at most once per session. The entry point `app.py` configures the page, checks whether Supabase authentication is enabled, and dispatches to the page renderer selected in the sidebar. Nine user-facing pages consume those primitives (Fig. 1). A small resolver (`src/profile.py`) compares the uploaded frame against the reference schema and decides which *data-profile* the session should run in. When the columns match the ecophysiology schema it keeps the physiology profile, with its domain defaults, presets and schema report; otherwise it falls back to a generic profile that drops those presets and the "missing required column" warnings, so an unrelated agronomic or biological table is analysed cleanly without any change to the code.

The `src/` package is organized by concern. `schema.py` declares a reference dictionary of 31 columns in three severity tiers (required, recommended, op-

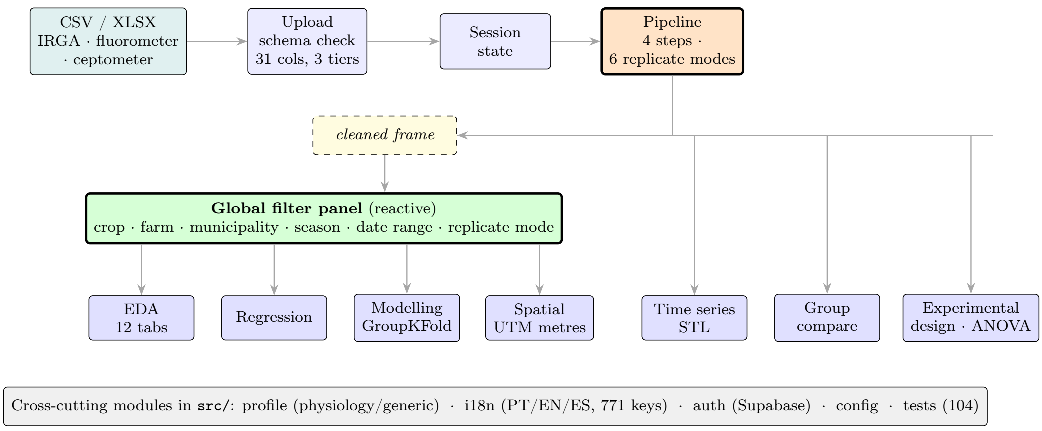


Figure 1: High-level data-flow architecture of *PhysioFlow*. Raw exports from IRGA, chlorophyll fluorometer, and ceptometer instruments are validated against the reference schema upon upload, cached in the Streamlit session state, and consumed by the Pipeline, which produces a single cleaned frame. A shared, reactive *global filter panel* (crop, farm, municipality, season, date range, and replicate mode) is injected at the top of the four analytical pages (EDA, Regression, Modeling, Spatial) and propagates instantly to every chart, test and model; the Time Series, Group-comparison and Experimental-design pages read the cleaned frame directly through their own controls. Library stack: Pipeline → `pandas`/`NumPy`; modelling → `scikit-learn` [23]; inference → `SciPy`/`statsmodels`; visualisation → `seaborn` [24, 25]; geospatial → `geopandas`, `pyproj`, `libpysal`/`esda` [26].

tional), each mapped to the dependent feature, and validates types and coordinate ranges without blocking the upload. `pipeline.py` hosts the deterministic cleaning primitives and a robust date-column detector that coerces heterogeneous Excel `object` columns to `datetime64`. `stats_utils.py` implements bias-corrected Cramér's V [27] and a categorical-confounding auditor, and it also holds the analysis-of-variance routines behind the experimental-design page: the split-plot, strip-plot, nested and ANCOVA models, and the Scott–Knott, Duncan, LSD, Scheffé and Dunnett mean-comparison procedures, each checked against its counterpart in R. Model estimators live in `ml/model_registry.py`, which declares two registries wrapped in a `scikit - learn` pipeline: five regressors (Linear Regression, Random Forest [28], Decision Tree, Histogram Gradient Boosting [29] and k-NN) and seven classifiers (Logistic Regression, Random Forest, Decision Tree, Histogram Gradient Boosting, k-NN, SVM and Naive Bayes); `profile.py` resolves the physiology/generic data-profile described above. The cross-cutting modules under `i18n/` (771 keys at full PT/EN/ES parity), `auth/`, `config/` and `tests/` (104 `pytest` functions) serve every page.

The implementation follows established software-engineering practices: public functions in `src/` carry type annotations; the 104-function suite covers the schema validator, the confounding utilities and the experimental-design routines (the latter cross-validated against R), measured with `pytest -cov`; `ruff` enforces style and `mypy` validates static types on the analytical core; and a GitHub Actions workflow runs the test, lint and type-check matrix on Python 3.10–3.12 for every push and pull request. New estimators, cleaning primitives and schema columns are added declaratively through the corresponding registries in `src/`, so the user interface extends itself without bespoke widget code.

2.2. Software functionalities

Upload (Fig. 2).. Accepts CSV, TSV, TXT (with a configurable delimiter), XLSX and XLS files. After parsing, the schema validator classifies every expected column as *required*, *recommended* or *optional*, checks data types, flags columns that exist but are 100% empty (distinct from "missing"), and verifies that latitude/longitude lie within plausible ranges. Missing columns disable dependent features without blocking the upload; when the dataset does not match the physiology schema, the application switches to its generic data profile.

Pipeline.. Four deterministic steps, text standardization, removal of rows without essential metadata, removal of empty grid points, and replicate treatment, followed by a tabular step-by-step report (Fig. 3) recording rows in/out and percentage removed at each step, with a warning whenever any step discards more than 50% of the rows. Replicates of chlorophyll and LAI can be handled in six modes: mean, *median* (robust to outliers; added at the suggestion of an external researcher), replicate-unfolding into independent rows, and exclusive selection of replicate 1, 2 or 3.

Exploratory Data Analysis.. Twelve tabs: descriptive statistics; a data-quality tab that audits categorical *confounding* via Cramér's V (Fig. 4); univariate distributions with KDE; boxplots by group; pair plots; Pearson, Spearman or Kendall correlation matrices; a lat/lon scatter map; temporal aggregation; categorical composition charts; non-parametric inference (Kruskal–Wallis [30] with a minimum-N-per-group slider, Shapiro–Wilk [31] accompanied by Q–Q plots, and VIF [32] with an explicit caveat that derived indices such as C_i/C_a, A/C_i and WUE inflate VIF by construction); a top-N hotspots ranking; and a multi-method outlier audit (Z-score, IQR, Isolation Forest [33], Local Outlier Factor [34], Elliptic Envelope) with a consensus criterion and an assumptions caption.

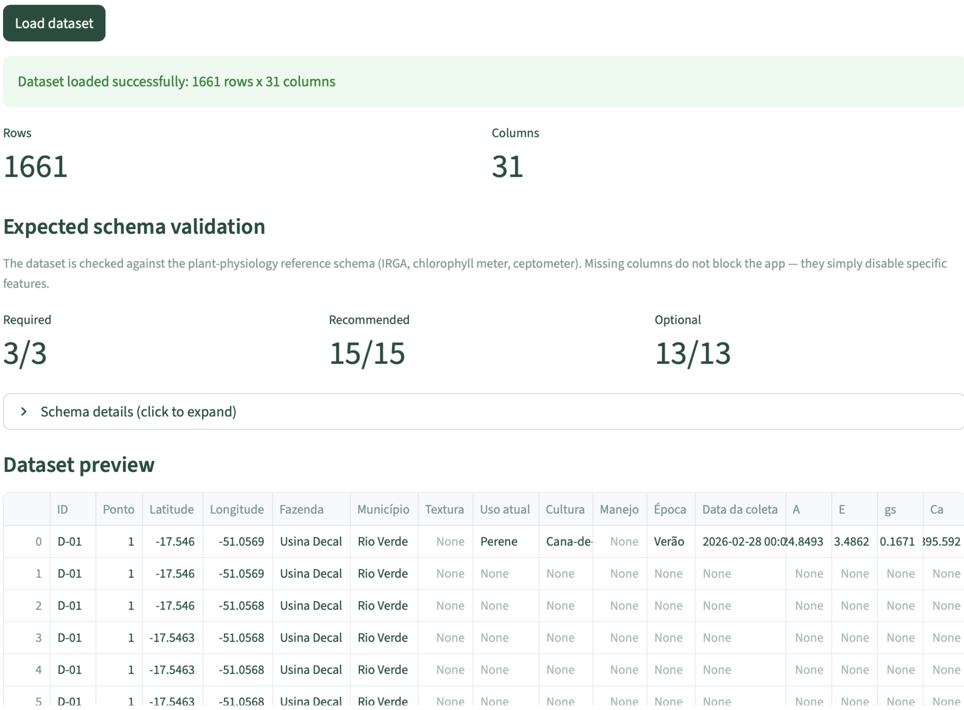
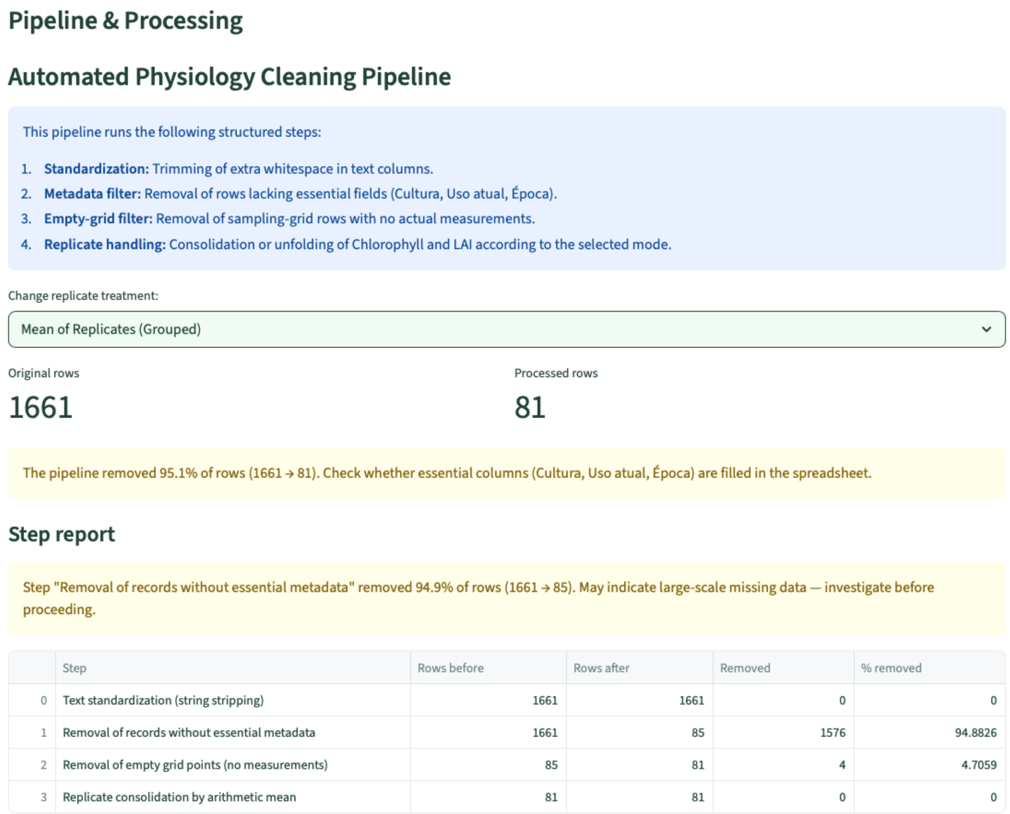


Figure 2: Upload page after ingestion of an ecophysiology dataset. The schema validator reports each expected column with its severity tier, the resolved column name, the detected type and the dependent feature.

Regression and Modeling.. The Regression page offers physiological bivariate presets (e.g. g_s vs. A, C_i vs. A) and free custom regression. On the Modeling page, the user first chooses the task. Regression compares five estimators and reports holdout R^2, MAE, RMSE and cross-validated R^2; classification compares seven and reports accuracy, precision, recall, F_1 and a confusion matrix. Either way, an *optional GroupKFold* with a grouping-column selector (by default a synthetic *farm + point* key) curbs pseudoreplication [16], and the folds are stratified when the task is classification.

Experimental Design.. A separate page (`./experimental_design.py`) treats the uploaded table as a designed experiment. The user names a categorical treatment factor and a numeric response, then chooses the layout. The options run from completely randomized (CRD) and randomized complete block (RCBD) through two- and three-factor factorials, with or without blocks, and the Latin square, up to the compound-error designs: split-plot, strip-plot and nested; analysis of covariance (ANCOVA) and dose-response regression sit alongside them. The page fits the model, prints the ANOVA table and checks the residuals on screen with Shapiro–Wilk and Levene tests



Pipeline & Processing

Automated Physiology Cleaning Pipeline

This pipeline runs the following structured steps:

1. **Standardization:** Trimming of extra whitespace in text columns.
2. **Metadata filter:** Removal of rows lacking essential fields (Cultura, Uso atual, Época).
3. **Empty-grid filter:** Removal of sampling-grid rows with no actual measurements.
4. **Replicate handling:** Consolidation or unfolding of Chlorophyll and LAI according to the selected mode.

Change replicate treatment:

Mean of Replicates (Grouped)

Original rows

1661

Processed rows

81

The pipeline removed 95.1% of rows (1661 → 81). Check whether essential columns (Cultura, Uso atual, Época) are filled in the spreadsheet.

Step report

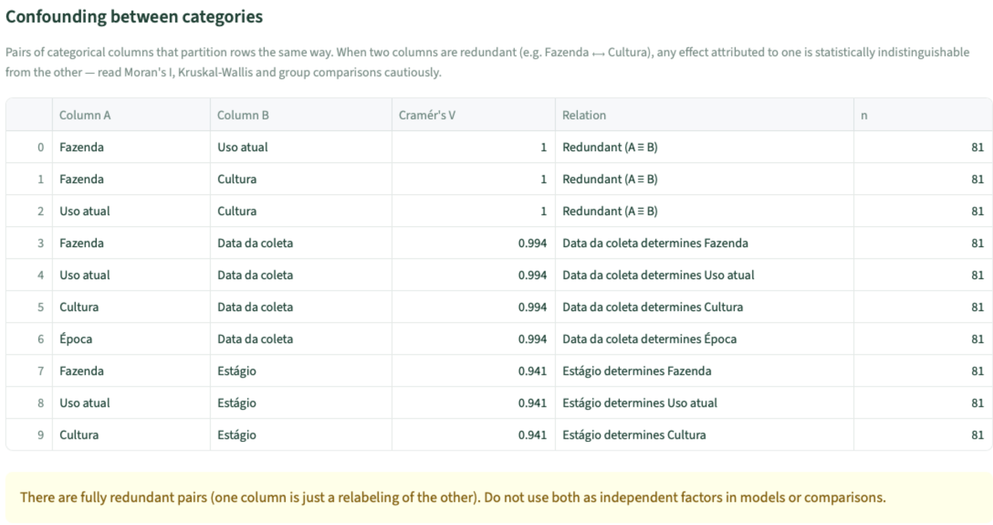
Step "Removal of records without essential metadata" removed 94.9% of rows (1661 → 85). May indicate large-scale missing data — investigate before proceeding.

	Step	Rows before	Rows after	Removed	% removed
0	Text standardization (string stripping)	1661	1661	0	0
1	Removal of records without essential metadata	1661	85	1576	94.8826
2	Removal of empty grid points (no measurements)	85	81	4	4.7059
3	Replicate consolidation by arithmetic mean	81	81	0	0

Figure 3: Step-by-step audit of the cleaning pipeline applied to the real Rio Verde dataset. Each row reports lines before/after and the share removed; the essential-metadata filter alone removes 94.9% of the raw rows, triggering the high-discard warning.

next to Q–Q and residual-versus-fitted plots, and it warns when an assumption fails, when a group is too thin in levels or replicates, or when there are simply too many levels to read. For the mean comparison, the user picks among Tukey HSD, Scott–Knott [35], Duncan, LSD/Fisher, Scheffé, and Dunnett (each treatment against a chosen control), reported as compact-letter displays. A reproducible R/Python snippet accompanies every result, and we validated each procedure against its reference R implementation, among them the official `ScottKnott` package and the ASReml cookbook datasets.

Spatial Analysis.. IDW interpolation, global Moran's I and LISA cluster maps [36], Getis–Ord G_i^* [37], UTM grid aggregation, ordinary kriging [38] with a fitted spherical variogram, and an optional municipality basemap. All distance-based computations are performed in *UTM metres* with an



Confounding between categories

Pairs of categorical columns that partition rows the same way. When two columns are redundant (e.g. Fazenda ⟷ Cultura), any effect attributed to one is statistically indistinguishable from the other — read Moran's I, Kruskal-Wallis and group comparisons cautiously.

	Column A	Column B	Cramér's V	Relation	n
0	Fazenda	Uso atual	1	Redundant (A ≡ B)	81
1	Fazenda	Cultura	1	Redundant (A ≡ B)	81
2	Uso atual	Cultura	1	Redundant (A ≡ B)	81
3	Fazenda	Data da coleta	0.994	Data da coleta determines Fazenda	81
4	Uso atual	Data da coleta	0.994	Data da coleta determines Uso atual	81
5	Cultura	Data da coleta	0.994	Data da coleta determines Cultura	81
6	Época	Data da coleta	0.994	Data da coleta determines Época	81
7	Fazenda	Estágio	0.941	Estágio determines Fazenda	81
8	Uso atual	Estágio	0.941	Estágio determines Uso atual	81
9	Cultura	Estágio	0.941	Estágio determines Cultura	81

There are fully redundant pairs (one column is just a relabeling of the other). Do not use both as independent factors in models or comparisons.

Figure 4: Data-quality tab auditing categorical confounding. On the Rio Verde dataset the tool detects that *Fazenda* (farm), *Cultura* (crop) and *Uso atual* (land use) are perfectly redundant (Cramér's V = 1.00), warning the user that between-group contrasts on these fields are statistically unidentifiable.

EPSG code derived automatically from the mean longitude, correcting the anisotropy of degree-based latitude/longitude distances.

Time Series.. Daily resampling (mean or median) and STL decomposition [39], *blocked when fewer than ten distinct dates are present* so that sparse campaigns do not yield a decomposition dominated by interpolation.

Two-group comparison and Authentication.. A comparative page partitions the dataset via manual multi-select or a case-insensitive substring pattern and returns a Mann–Whitney U test [40] per variable, paired log-linear regressions, and an hourly profile. An optional Supabase login (disabled by default) adds email/password authentication; the application is fully functional without configuration.

3. Illustrative example

3.1. A crop-ecophysiology campaign

A real ecophysiology campaign in Rio Verde, GO (Brazilian Cerrado), comprising 1 661 raw measurement rows across soybean and sugarcane fields, was ingested via the Upload page. The schema validator confirmed the required

metadata and physiological variables and flagged the *Manejo* (soil management) and *Textura* (soil texture) columns as present but 100% empty.
The Pipeline was run in the default mean-replicate mode. Its audit trail (Fig. 3) traces 1 661 input rows down to 81 sampling-point records: the essential-metadata filter alone removes 94.9% of the raw rows (an incomplete sampling grid in which most planned points carry no measurement), triggering the high-discard warning that alerts the user before any inference is attempted.
The cleaned frame was then submitted to the EDA data-quality tab (Fig. 4). The Cramér's V auditor reported that *Fazenda*, *Cultura* and *Uso atual* are mutually redundant (Cramér's V = 1.00): the two surviving farms map one-to-one onto the two crops (Usina Decal ≡ sugarcane, 27 records; Reunidas Baumgart ≡ soybean, 54 records). The guard-rail makes explicit that any nominally "between-farm" contrast is really an unidentifiable "between-crop" contrast, the kind of confounding that a plain grouped comparison, or a high spatial-autocorrelation statistic, would quietly read as spatial structure instead.
The Modelling page then predicted net photosynthesis (A, $n = 71$) from gas-exchange, fluorescence and chlorophyll predictors, comparing five supervised regressors in a single run under GroupKFold cross-validation keyed on the synthetic *farm + point* group to curb pseudoreplication (Fig. 5); Linear Regression attained the highest cross-validated skill (CV $R^2 = 0.95$), with the tree ensembles close behind. Finally, the Time Series page was invoked: with only three distinct collection dates (2025-12-19 to 2026-02-28), the STL guardrail *blocked* the decomposition and explained why (Fig. 6), rather than returning a seasonal component that would be 90% interpolation.
End-to-end, the four-step pipeline on the 1 661-row dataset completes in under two seconds on a 16 GB-RAM laptop (Python 3.10, pandas 2.0), and each EDA tab, model fit and spatial computation renders interactively without leaving the browser.

3.2. Datasets beyond ecophysiology

To show that the platform is not bound to its original domain, we ran two public datasets through the same interface; in both, the profile resolver found no match to the ecophysiology schema and silently switched to the generic profile, so neither analysis required a line of code or any change to the application.
The first is the classical Yates oats trial [41], three oat varieties under four nitrogen rates across six blocks (72 rows), uploaded as a plain CSV and analysed on the Experimental-design page as a split-plot, with variety as the whole-plot factor and nitrogen as the sub-plot factor. The resulting

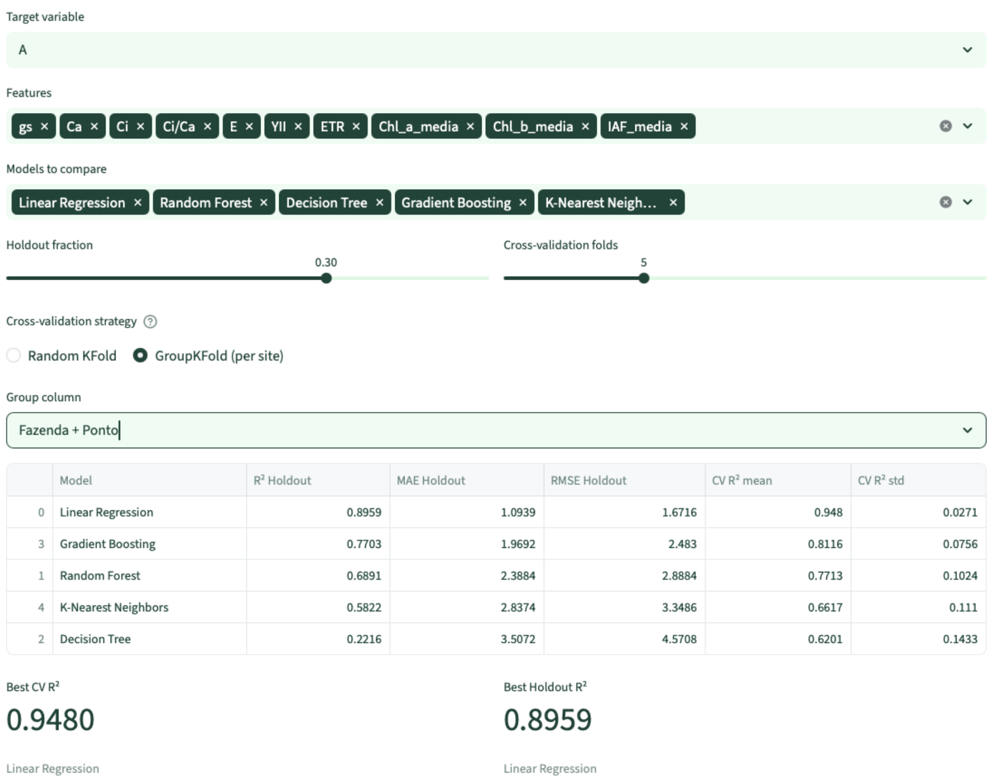


Figure 5: Modelling page comparing five supervised regressors for net photosynthesis (A) on the Rio Verde dataset in a single run, reporting holdout R^2, MAE and RMSE alongside cross-validated R^2 (mean and standard deviation). Linear Regression attains the highest cross-validated skill, with the tree ensembles close behind; here the GroupKFold-per-site guard-rail (grouping on *farm* + *point*) is enabled to curb pseudoreplication.

ANOVA reproduces the reference fit from `nlme::Oats` [42] to the decimal in every stratum (Table 2): the whole-plot variety effect is non-significant, the sub-plot nitrogen effect is strong, and the variety×nitrogen interaction is negligible. The post-hoc Tukey test on nitrogen recovers the textbook dose response, yield saturating between the two highest rates, and the page's residual checks (Shapiro–Wilk, Levene) raise no flag.

The second is the Palmer penguins survey [43], a biology dataset of 344 morphometric records from three species, analysed on the EDA and Modelling pages. The EDA recovered the known flipper-length/body-mass correlation ($r = 0.87$) and the sign reversal of bill depth against mass that is a textbook instance of Simpson's paradox, while the Modelling page in classification mode separated the three species almost perfectly: Random Forest, SVM and k-NN at 1.00 holdout accuracy and 0.99 under stratified cross-validation,

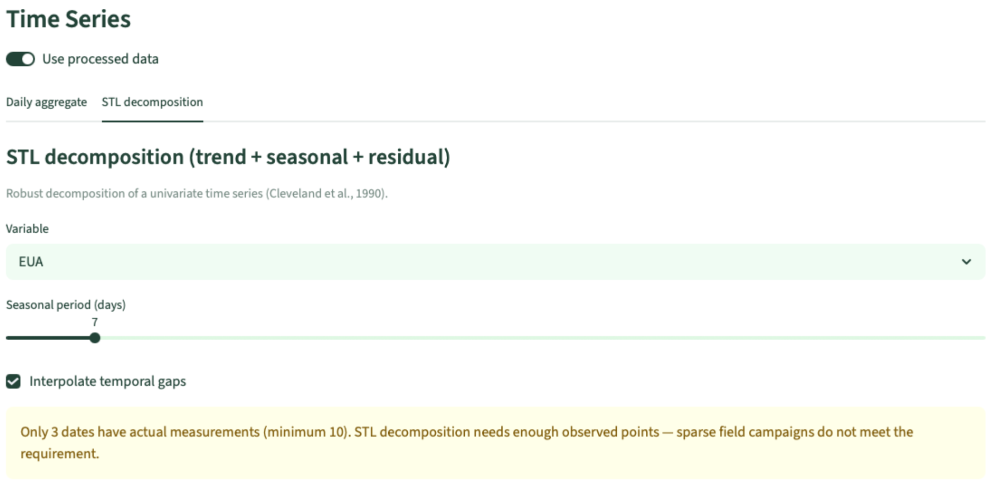


Figure 6: Time Series page on the Rio Verde dataset. Because the campaign contains only three distinct collection dates, below the ten-date threshold, the STL guard-rail blocks the decomposition and reports the reason instead of producing an interpolation-dominated result.

Term (stratum)	**df**	**PhysioFlow** F (p)	`nlme::Oats` F (p)
Variety (whole-plot)	2, 10	1.485 (0.272)	1.485 (0.272)
Nitrogen (sub-plot)	3, 45	37.686 ($< 10^{-11}$)	37.686 (< 0.001)
Variety × Nitrogen	6, 45	0.303 (0.932)	0.303 (0.932)

Table 2: Split-plot ANOVA of the Yates oats trial computed by PhysioFlow's experimental-design module against the R `nlme::Oats` reference. The F statistics match to the decimal in all three strata.

matching the strongest published benchmarks for this dataset. A split-plot agronomy trial and a multi-species biology survey thus run through the same guard-railed interface that served the Cerrado gas-exchange campaign.

4. Impact

PhysioFlow brings the cleaning, exploratory analysis, supervised modeling, and formal experimental-design analysis needed to interpret crop-ecophysiology data into a single interactive platform, closing a gap left by instrument manufacturers, who provide no free or open-source tools for these stages. For agronomists, plant scientists, and graduate students without a programming background, especially in developing countries and under-resourced institutions, this barrier often delays the analysis of campaigns that were already expensive to run in terms of equipment, logistics, and labor.

PhysioFlow's main contribution is methodological, not just operational: its statistical guardrails turn assumptions that researchers usually check informally or skip into explicit on-screen warnings. The Cramér's V confounding auditor, the VIF caveat for derived physiological indices, the minimum-N filter for Kruskal–Wallis, the Q–Q plots beside normality tests, the optional GroupKFold against pseudoreplication, the residual normality and homoscedasticity checks on every experimental-design fit, and the sparsity block on STL each warn a non-statistician user before a familiar mistake is made. In the Rio Verde campaign they caught a perfectly confounded design and a series too short for seasonal decomposition at once, problems that, left unflagged, would more likely have come up only at peer review.
PhysioFlow is currently used within the Goiás Verde initiative (Instituto Federal Goiano and CEAGRE) to analyse the photosynthetic performance of soybean and sugarcane across cropping systems of the Brazilian Cerrado, and it supports the data-analysis training of several graduate students and their supervisees in the associated post-graduate programme, who use the application both for their own datasets and for teaching. An earlier sibling of the platform, targeting soil greenhouse-gas fluxes, was published and presented to the agronomy and remote-sensing community at the 1st International Symposium on Climate and Agricultural Resilience / 5th CEAGRE Agro Experience, where its four core analytical modules were also delivered as a hands-on training workshop [18]. Because it is open-source, free of licensing fees, and dataset-agnostic, the tool can be reused for any tabular research dataset that needs diagnostic-driven cleaning, EDA, and supervised modeling; we see immediate scope in digital agriculture, plant phenotyping, and environmental monitoring. Its trilingual (PT/EN/ES) interface also widens its reach across the Ibero-American research community. The current release (v1.0) comprises approximately 7,400 lines of Python across the `src/` package, 104 `pytest` unit tests, and 771 user-interface keys at full PT/EN/ES parity. Early feedback from collaborating laboratories indicates that analyses that used to take several days of spreadsheet work and ad hoc scripting now finish in a few hours within the interface, leaving researchers more time for the biology behind the numbers.

5. Conclusions

We presented *PhysioFlow*, an open-source Python/Streamlit interactive platform for crop-ecophysiology datasets produced by IRGA, chlorophyll-meter and ceptometer instruments. In one trilingual interface it brings together ingestion and schema validation, a configurable six-mode replicate pipeline, exploratory analysis with statistical guard-rails, multi-method outlier audit-

ing, bivariate and machine-learning regression and classification under an optional GroupKFold, geospatial analysis in UTM metres (IDW, Moran's I/LISA, Getis–Ord G_i^* and ordinary kriging) and STL time-series decomposition behind a sparsity block. To these it now adds a formal experimental-design module, CRD, RCBD, factorial, Latin square, split-plot, strip-plot and nested designs, with ANCOVA and dose-response, whose assumption checks and post-hoc comparisons we validated against R. An automatic physiology/generic data-profile carries the tool beyond ecophysiology, so it reads any tabular agronomic or biological dataset.
Its distinguishing feature is the integration of contextual statistical guardrails directly into the user interface, helping non-statistician users avoid the pitfalls of confounding, multicollinearity, pseudoreplication, and sparsity common in ecophysiological field data, as demonstrated in a real Cerrado campaign.
Planned developments include physiological range validation in the schema (`valid_range` per column), integration of the instrument data dictionary as in-app tooltips, raising test coverage above 70% across the `ml/`, `state` and `auth` modules, a quasi-constant-variable warning in the correlation panel, and a public Docker image for one-command deployment. On the analytical side, automatic fitting of physiological response curves (e.g. A/C_i and light-response curves), automated report export, and predictive guard-rails that flag physiological-stress patterns to support decision-making are planned as future directions.

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